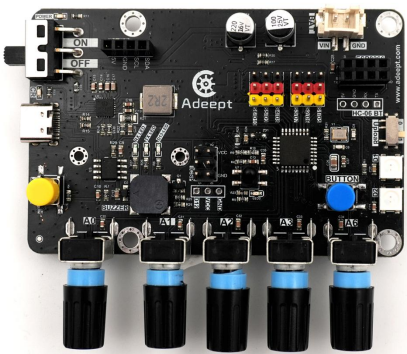


Lesson 9 Control the LED via Button

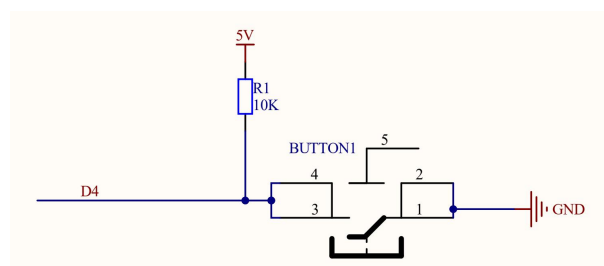
9.1 Overview

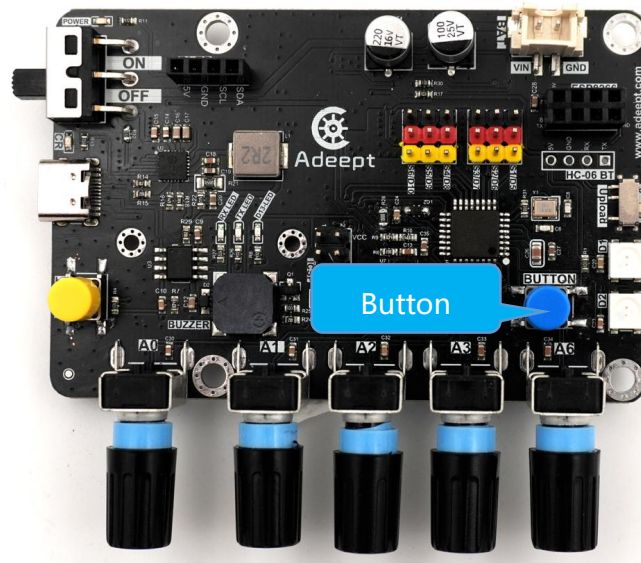
In this lesson, we will introduce the buttons on the control board and learn how to use buttons through an example of controlling LEDs with buttons.

9.2 Required Components

Components	Quantity	Picture
Adept Robotic Arm Control Board	1	
Type-C USB Cable	1	

9.4 Wiring Diagram





PINS of AdeepT Robotic Arm Control Board	Button
D4	GND
VCC	VCC

9.5 Demonstration

1. Connect your computer and AdeepT Robotic Arm Control Board with a USB cable.
2. Open **"09_Button"** folder in **"/Code"** , double-click **"09_Button.ino"** .

```

09_Button | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Select Board

09_Button.ino
1  const int buttonPin = 4;  // Button connected to pin D4
2  const int ledPin = 13;    // Default pin 13 for Uno
3
4  void setup() {
5      Serial.begin(115200);
6      pinMode(buttonPin, INPUT_PULLUP);
7      pinMode(ledPin, OUTPUT);
8      Serial.println("Button test started, press or release the button on pin D4");
9  }
10
11 void loop() {
12     int buttonState = digitalRead(buttonPin);
13     // Internal pull-up mode: LOW when pressed, HIGH when released
14     if (buttonState == LOW) {
15         digitalWrite(ledPin, LOW); // Turn on built-in LED when pressed
16         Serial.println("Status: Pressed");
17     } else {
18         digitalWrite(ledPin, HIGH); // Turn off built-in LED when released
19         Serial.println("Status: Released");
20     }
21     delay(100);
22 }

```

3. Select development board and serial port.

Board: **Tools---**>**Board---**>**Arduino AVR Boards---**>**Arduino Uno**

Port: **Tools --->Port---**>**COMx**

Note: The port number will be different in different computers.



4. After opening, click  to upload the code program to the Arduino. If there is no error warning in the console below, it means that the Upload is successful.

```
Output
Sketch uses 2952 bytes (9%) of program storage space. Maximum is 32256 bytes.
Global variables use 82 bytes (4%) of dynamic memory, leaving 1966 bytes for local variables. Maximum is 2048 bytes.
```

5. After the program runs successfully, the LED will light up when you press the button and turn off when you release it.

9.6 Code

```
01 const int buttonPin = 4;    // Button connected to pin D4
02 const int ledPin = 13;      // Default pin 13 for Uno
03
04 void setup() {
05     Serial.begin(115200);
06     pinMode(buttonPin, INPUT_PULLUP);
07     pinMode(ledPin, OUTPUT);
08     Serial.println("Button test started, press or release the button on pin D4");
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10
11 void loop() {
12     int buttonState = digitalRead(buttonPin);
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17     } else {
18         digitalWrite(ledPin, HIGH); // Turn off built-in LED when released
19         Serial.println("Status: Released");
20     }
21     delay(100);
22 }
```